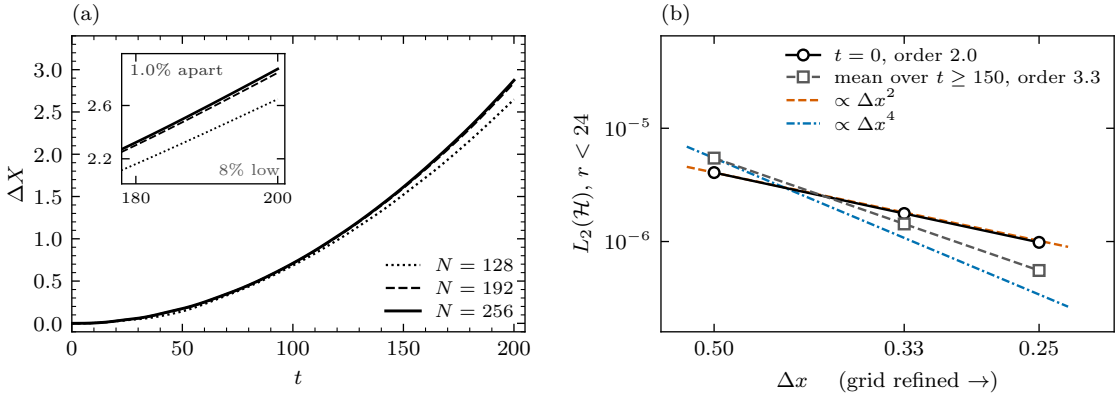

(a)
ΔX
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
50
100
150
200
t
1.0% apart
2.6
2.2
180
200
8% low
$N = 128$
$N = 192$
$N = 256$
(b)
$L_2(\mathcal{H})$, $r < 24$
10^{-5}
10^{-6}
$t = 0$, order 2.0
mean over $t \geq 150$, order 3.3
$\propto \Delta x^2$
$\propto \Delta x^4$
0.50
0.33
0.25
Δx (grid refined $\rightarrow$)